KGiSL Institute of Technology

(An Autonomous Institution)

Affiliated to Anna University, Approved by AICTE, Recognized by UGC,
Accredited by NAAC & NBA (B.E-CSE,B.E-ECE, B.Tech-IT),
365, KGiSL Campus, Thudiyalur Road, Saravanampatti, Coimbatore – 641035.

SMART ATTENDANCE PREDICTER

A SUMMER INTERNSHIP REPORT

Submitted by

GOWDHAM S (711724UAD133)

in partial fulfillment for the award of the degree

of

BACHELOR OF TECHNOLOGY

IN

DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

KGiSL INSTITUTE OF TECHNOLOGY

JULY 2026

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BONAFIDE CERTIFICATE

Certified that this Internship Report on **Smart Attendance Predictor Internship** at **Litz tech** is the bonafide work carried out by **GOWDHAM S** , who belongs to III Year Artificial Intelligence and Data Science "A" during V Semester of Academic Year 2026-2027.

FACULTY INCHARGE **HEAD OF THE DEPARTMENT**

Certified that the candidate was examined by us for the Summer Internship Viva held on _____at KGiSL Institute of Technology, Saravanampatti, Coimbatore 641035.

INTERNAL EXAMINER **EXTERNAL EXAMINER**

ACKNOWLEDGEMENT

We express our deepest gratitude to our **Chairman and Managing Trustee Dr. Ashok Bakthavathsalam** for providing us with an environment to complete our Internship project successfully.

We are extremely grateful to our **CEO of Academic Initiatives, Mr. Aravind Kumar Rajendran.** We extend our heartfelt thanks to our honorable **Principal, Dr. Suresh Kumar S**, for his support, guidance, and blessings. We also express our sincere gratitude to our beloved **Dean (Academic and Research), Dr. Kannan**, for his encouragement and support.

We would like to thank **Sathish R , Head of the Department**, and **Internship Coordinator Ms. Jasmine Saranya** , Department of Artificial Intelligence and Data Science for firm support during the entire course of this Internship and who modeled us both technically and morally for achieving greater success in this project work.

We also thank all the faculty members of our department for their help in making this Internship project a successful one. Finally, we take this opportunity to extend our deep appreciation to our Family and Friends, for all they meant to us during the crucial times of the completion of our project.

INTERNSHIP

OFFER LETTER

LITZ Tech

ACCEPTANCE LETTER

To
KGISL INSTITUTE OF TECHNOLOGY

Dear Sir/Madam,

Sub: Acceptance of Internship - Reg.

We are pleased to grant permission for **Mr Gowdham S (711724UAD133)** of the **B.Tech AI & DS** department from **KGISL Institute of Technology** to pursue his **Internship** in our organization, for the duration from **June 1st, 2026** to **June 15th, 2026**.

During this period, the participant will be provided with the necessary working platform and resources required for the successful completion of the training.

Thanking you,

Student ID: LITZS2846
Document ID: LITZC2846

Authorized Signatory

INTERNSHIP

CERTIFICATE

LITZ Tech

CERTIFICATE OF INTERNSHIP

This certifies that **Mr Gowdham S (711724UAD133)** from **KGISL Institute of Technology** has successfully completed his **Artificial Intelligence Internship** at **Litz Tech**, from **2026-06-01** to **2026-06-15**.

A high degree of dedication and professionalism was demonstrated during this time by fully executing the assigned responsibilities. Throughout the **Internship**, the candidate maintained an excellent standard of attendance and punctuality. We deeply appreciate his hard work and wish him success in all future endeavors.

Authorized Signatory

Student ID: LITZS3005
Certificate ID: LITZC3005

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ABSTRACT

The internship at Litz Tech provided me with practical, industry-oriented exposure to full-stack AI-assisted web application development by designing and developing **Smart Attendance Predictor**, a modern web platform that records student attendance and predicts future attendance trends and attendance-shortage risk. The project focused on building a responsive, production-quality system that helps institutions monitor attendance, identify students at risk of falling below the required percentage, and take timely action.

Throughout the internship, I strengthened my knowledge of frontend component design, REST API development, JWT-based authentication, relational database modelling, CRUD operations, data preprocessing, machine learning–based prediction, and data visualization. I learned how to structure a full-stack project, connect a frontend application to a Python backend, validate and secure user data, and implement explainable prediction features such as attendance percentage forecasting and low-attendance alerts.

This internship enhanced my problem-solving abilities, logical thinking, debugging skills, and full-stack development knowledge while providing practical experience in developing a real-world, deployable web application.

INTRODUCTION

The internship served as an excellent opportunity to bridge the gap between theoretical knowledge and practical software development. During this internship, I designed and developed **Smart Attendance Predictor**, an AI-powered attendance management and prediction system built as a full-stack web application.

Manual attendance tracking is time-consuming and does not warn students or faculty before attendance falls below the mandatory threshold. Smart Attendance Predictor addresses this by storing daily attendance records, calculating attendance percentages automatically, and using a prediction model to forecast whether a student is likely to meet the required percentage by the end of the semester.

The frontend was developed using React with Tailwind CSS to provide a clean, dashboard-style interface. The backend was developed using Python (FastAPI) with SQLAlchemy as the ORM and SQLite as the database, exposing a REST API consumed by the frontend. The prediction module was built using Python libraries such as Pandas and Scikit-learn. JSON Web Tokens (JWT) were used to secure authentication and protect routes.

OBJECTIVES

- To understand the fundamentals of full-stack web application development.
- To design and develop a user-friendly, AI-powered attendance management system.
- To build a responsive graphical user interface using React and Tailwind CSS.
- To implement secure authentication using JWT tokens.
- To implement student management and daily attendance marking functionality.
- To calculate attendance percentage automatically for each student and subject.
- To predict future attendance percentage and identify students at risk of attendance shortage.
- To connect the application with a SQLite database using SQLAlchemy.
- To perform CRUD operations for attendance records through a REST API.
- To visualize attendance analytics using interactive charts.
- To generate alerts and downloadable reports for low-attendance students.
- To improve logical thinking, debugging skills, and full-stack development knowledge.
- To gain practical experience in developing a real-world software application.

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TECHNOLOGIES USED

CATEGORY	TECHNOLOGY	USED FOR
Front end	React + Vite	Building the user interface
Styling	Tailwind CSS	Making the design clen and responsive
Charts	Recharts	Showing attendance graps
Back end	Python	Handling request business logic
Database	SQlite	Storing sudents and attendance data
Prediction	Pandas	Prediciting future attendance
Security	JWT + Password hasing	Secure login and protected pages
Testing	Browser Devtool	Testing APIs and fixing bugs

METHODOLOGY & TASKS

The internship followed a structured, task-based development methodology, where each stage focused on a specific development activity. The process started with understanding the requirements of an attendance management system and identifying the major modules required for tracking and predicting attendance.

The application was then developed using React for the frontend user interface and FastAPI with SQLAlchemy/SQLite for the backend. The prediction model was trained and tested separately and then integrated with the API. Each module was developed and tested individually before integrating it into the complete application

Project Component: Smart Attendance Predictor

Description:

Smart Attendance Predictor is a full-stack, AI-assisted web application designed to simplify attendance tracking and help prevent attendance shortage. The system allows faculty to mark and manage attendance through a responsive dashboard, while students and administrators can view attendance percentages, trends, and predicted outcomes.

The application uses SQLite (via SQLAlchemy ORM) to store data securely and exposes REST API endpoints that provide CRUD operations such as adding, viewing, updating, and deleting records.

Tasks Performed:

1. Authentication Module

Developed a secure Register/Login interface that allows authorized users to access the platform. The module provides:

- User registration and login interface
- Password hashing and JWT token-based session validation
- Error messages for invalid login credentials
- Protected routes that redirect unauthenticated users to the login page

2. Dashboard Module

The dashboard acts as the main interface after successful login. It provides:

- Stat cards summarizing total students, average attendance, students at risk, and today's absentees
- Attendance trend and subject-wise attendance charts
- Predicted attendance overview
- Recent activity feed

3. Student Management Module

This module is used to maintain student profiles. It allows users to:

- Add, view, update, and delete student records (name, register number, department, year)
- Search and filter students by department or year
- View an individual student's attendance history and percentage

4. Attendance Marking Module

This module handles daily attendance entry. Faculty can:

- Select a class, subject, and date
- Mark students as Present, Absent, or On-Duty
- Edit previously submitted attendance
- Automatically update attendance percentage after every entry

5. Attendance Prediction Module

The prediction module uses historical attendance data to forecast future attendance. It provides:

- Predicted end-of-semester attendance percentage for each student

- Classification of students into Safe, Warning, and At-Risk categories
- Number of classes a student must attend to reach the required percentage
- Model confidence indicators for predicted values

6. Alerts & Notifications Module

This module surfaces low-attendance warnings and at-risk alerts, with the ability to mark notifications as read.

7. Reports & Analytics Module

The Reports module visualizes monthly attendance trends, subject-wise and department-wise attendance, and risk distribution charts, with CSV export support.

8. Database Management

SQLite (via SQLAlchemy ORM) was used as the database. It stores information related to Users, Students, Subjects, Attendance Records, Notifications, and precomputed prediction results. REST API endpoints built with FastAPI perform database operations such as:

- POST – create records (students, attendance entries)
- GET – retrieve and filter records
- PUT/PATCH – update records (e.g., edit attendance, update student details)
- DELETE – remove records

9. User Interface Design

The graphical user interface was developed using React with a component-based architecture (Sidebar, Navbar, Layout, Card, Button, Input, Badge, Table). The interface was designed to provide a simple, responsive, and user-friendly experience across devices

DEVELOPMENT PHASES

Phase 1: Requirement Analysis – Identified the requirements and major features of the Smart Attendance Predictor.

Phase 2: Database Design – Designed the SQLAlchemy models and created tables required for users, students, subjects, attendance records, and notifications.

Phase 3: Backend API Development – Built the FastAPI REST API with routers for authentication, students, attendance, predictions, and notifications, along with JWT-based security.

Phase 4: Frontend UI Design – Designed the login/register pages, dashboard, and layout components using React and Tailwind CSS.

Phase 5: Student and Attendance Module Development – Implemented CRUD operations and UI pages for managing students and marking attendance.

Phase 6: Data Preparation and Model Training – Cleaned and prepared attendance data, engineered features (attendance rate, absence streaks, weekday patterns), and trained the prediction model.

Phase 7: Prediction, Alerts and Analytics – Integrated the model with the API, implemented risk classification, alerts, and analytics dashboards with charts.

Phase 8: Integration, Testing and Debugging – Connected the React frontend to the FastAPI backend, tested each module individually, identified and corrected errors, and verified the complete workflow from login to prediction.

OUTPUT

TASK 1: TASK 1: Login / Register Page

Output:

TASK 2: DASHBOARD

TASK 3: : Student Management

TASK 4: Attendance Marking

TASK 5: Attendance Prediction

TASK 6: Alerts & Notifications

TASK 7: Reports & Analytics

CONCLUSION

The internship at Litz Tech was a valuable learning experience that helped me understand the practical aspects of full-stack, AI-assisted web application development. Through the Smart Attendance Predictor project, I successfully applied React, Python, FastAPI, and machine learning concepts to develop a modern, user-friendly web platform for tracking and predicting student attendance.

The project enhanced my understanding of frontend component design, REST API development, JWT authentication, database management using SQLAlchemy, CRUD operations, data preprocessing, model training, data visualization, and debugging techniques. I gained practical experience in developing modules such as authentication, dashboard, student management, attendance marking, prediction, alerts, and analytics.

The development of this project improved my logical thinking, problem-solving abilities, database management skills, and full-stack programming knowledge. I also learned how to organize a full-stack project, connect a React application with a FastAPI backend, handle user inputs securely, and test different functionalities end to end. Overall, the internship strengthened my technical skills and gave me confidence in developing real-world, AI-assisted web applications.

FUTURE ENHANCEMENTS

The mobile application of the **Students Internship Management Portal** can be further enhanced with additional features to improve functionality and user experience.

- Add push notifications for internship updates and important announcements.
- Implement real-time notifications for application status changes.
- Add offline data support for selected application features.
- Enhance the mobile dashboard with internship statistics and analytics.
- Improve UI animations and interactive components for a better user experience.
- Add advanced search and filtering options for internship opportunities.
- Implement document upload and management features within the mobile application.
- Add in-app communication features for students and internship coordinators.
- Improve mobile application security and authentication mechanisms.
- Expand the Flutter application with additional internship management functionalities.
- Optimize API communication and application performance.
- Publish the application for Android and iOS platforms.
- Implement document upload and management features within the mobile application.
- Add in-app communication features for students and internship coordinators.
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